

Steel Tongue Drum Capstone

Instrument-Maker Print Packet

Build packet folder: /mnt/c/Users/Tony/Documents/GitHub/steel-tongue-drum

Generated: 2026-05-08

This packet is the printable companion to the build folder. Take it shopping or into the shop. Tear sheets at page breaks.

File Map

File	Purpose
design.md	Project intent, catalog metadata, assumptions, and validation plan.
bom.csv	Starter bill of materials with part categories, quantities, drawing refs, and notes.
sourcing.csv	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
cut-list.csv	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
drawing-brief.md	Manufacturing drawing and technical product sketch brief.
assembly-manual.md	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
validation.csv	Target/measured values, tolerance, environment, result, and tuning/build action log.
supplier-rfq.md	Supplier email/request-for-quote starter.
visual-bom-brief.md	Art direction for an image-forward visual BOM.
wolfram-starter.wl	Wolfram starter for physics, optimization, visualization, and validation.
README.md	Project artifact.
family-spec.csv	Project artifact.
jig-decision.md	Project artifact.
photo-shotlist.md	Project artifact.
resources.md	Project artifact.
risks.md	Project artifact.
validation-report.md	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Steel Tongue Drum Design

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Intent

Create a traceable root-mode packet for a tuned steel tongue drum. The packet documents the first-pass geometry from `steel-tongue-drum-design-table.xlsx`, the manufacturing assumptions that can move pitch, and the validation workflow needed to turn estimates into a tuned instrument.

This design is intentionally conservative about claims. The workbook formulas size tongues from a cantilever model, but the final instrument must be tuned by measured cutting and strike tests.

Source Artifacts

Source	Role	Status
---	---	---
`steel-tongue-drum-design-table.xlsx`	Design table and first-pass geometry	Present
`README.md`	Root repo overview	Expanded in this packet
`instrument-maker-v4` references	Physics, sourcing, print packet, validator	Applied
Measured prototype data	Required for final tuning confidence	Not present

Workbook Inputs Used

Parameter	Value	Unit	Source / confidence
---	---	---	---
Shell outside diameter	12.000	in	workbook input, assumption
Shell height	6.000	in	workbook input, assumption
Steel thickness	0.079	in	workbook input, verify with calipers
Kerf width	0.040	in	workbook input, process-dependent
Tongue count	11	count	workbook note rows include bass row plus 10 numbered rows
Tongue width	1.000	in	workbook input, derived estimate
Port diameter	2.500	in	workbook input, derived estimate
Tuning standard	440	Hz	A4 reference
Selected K constant	32098	imperial	workbook steel table, first-pass only

Governing Model

Use the cantilever beam model only as an initial layout estimate:

$$f = K_{\text{steel}} * t / L^2$$

$$L = \sqrt{K_{\text{steel}} * t / f}$$

Where:

- `f` is target frequency in Hz.
- `K\_steel` is the workbook steel constant, currently 32098 for the selected steel row.
- `t` is steel thickness in inches.
- `L` is free tongue length in inches.

The model assumes an ideal fixed-free tongue. A real steel tongue drum departs from that ideal because the tongue remains part of a stressed plate/shell, is cut by a finite kerf, may be heat affected, may retain burrs, and is coupled to the vessel body and support system.

Target Tongue Table

The values below are first-pass estimates from the workbook formula family. Treat every length as a starting cut length with trim allowance, not a final tuned dimension.

Tongue	MIDI	Note	Target Hz	Derived length in	Derived length mm	Status
---:	---:	---	---:	---:	---:	---
Bass	50	D3	146.83	4.16	105.6	derived estimate
1	57	A3	220.00	3.39	86.2	derived estimate
2	58	A#3/Bb3	233.08	3.30	83.7	derived estimate
3	60	C4	261.63	3.11	79.0	derived estimate
4	62	D4	293.66	2.94	74.6	derived estimate
5	64	E4	329.63	2.77	70.4	derived estimate
6	65	F4	349.23	2.69	68.4	derived estimate
7	67	G4	392.00	2.54	64.5	derived estimate
8	69	A4	440.00	2.40	60.9	derived estimate
9	72	C5	523.25	2.20	55.9	derived estimate
10	74	D5	587.33	2.08	52.8	derived estimate

Shell / Helmholtz Estimate

The workbook includes a first-pass vessel estimate using a 12 in diameter, 6 in high partial volume assumption and a 2.5 in port. This is useful for checking whether the shell may reinforce or color the tongue field, but it should not be

treated as a final resonator design.

For this packet, the shell resonance is documented as a support variable:

- Verify actual internal volume after forming, welding, or purchased shell selection.
- Measure the port diameter after deburring and finish.
- Compare struck tongue spectra mounted and unmounted.
- Record whether final mounting shifts pitch or sustain.

Empirical Tuning Workflow

1. Rough-cut the tongues long or conservative, with rounded stress-relief termini.
2. Deburr both sides without thinning the root unpredictably.
3. Strike each tongue with the same mallet and support condition.
4. Log measured Hz and cents error in `validation.csv`.
5. Remove material in controlled passes, favoring tip/length correction first.
6. Re-measure after each tuning pass and after final mounting.
7. Re-check drift after finish cure and 24-72 hours of rest.

Manufacturing Assumptions

- Steel grade is not locked. Mild low-carbon sheet is the default planning case; stainless is acceptable only after revisiting K, cutting method, corrosion plan, and work hardening behavior.
- Thickness must be measured at the actual blank, not assumed from nominal sheet gauge.
- Laser cutting is fast but risks heat affected zones and local stress changes.
- Waterjet cutting reduces heat risk but can leave taper, abrasive edge texture, and local burrs.
- CNC milling can control slot geometry but needs serious fixturing and coolant/chip planning on thin steel.
- Every inside slot end needs a smooth radius to reduce crack initiation.

Unresolved Tuning Assumptions

- Actual steel grade, modulus, density, hardness, and residual stress.
- Actual kerf after the chosen cutting process.
- Whether the blank is flat sheet, preformed shell, salvaged vessel, or welded assembly.
- Final support/isolation scheme.
- Mallet hardness and strike point standard.
- Finish mass and corrosion-protection method.
- Whether the bass tongue and high tongues remain balanced in volume and sustain after final trimming.

Acceptance Criteria

The packet becomes public-review ready only after:

- Each tongue has target Hz, measured Hz, cents error, mallet condition, support condition, and environment logged.
- Rough-cut, deburr, first strike, tuning pass, final mount, and drift check rows are complete.
- At least one process note records actual kerf, burr condition, and heat/warp observations.
- `validation-report.md` is updated from measured data rather than assumptions.

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_id	part_name	qty	unit	required_spec	estimated_cost	cost_basis	critical_to_tuning	notes
BOM-001	Steel shell or formed blank	1	each	12 in diameter by 60 in high steel vessel	100	rough planing blank	Yes	Thick walls, refer to drawing for details. Do not use mild steel or aluminum.
BOM-002	Top tongue field blank	1	each	Integral shell top or welded steel top	35	rough planing 0.57 in thick	Yes	State origin of field. Avoid known hardened or heat-treated materials.
BOM-003	Rubber isolation feet	8	set	Low-profile rubber feet or grommet supports	8	rough planing 1.5 in thick	Yes	Hand finished. Avoid materials with pitch and sulfur.
BOM-004	Mallet pair	1	pair	Medium-soft rubber or yarn-wrapped mallets	15	rough planing 0.5 in thick	Yes	Head hardness must be recorded. Avoid materials that are too hard.
BOM-005	Edge trim or rim protection	1	length	Non-rattling edge trim compatible with finish	12	rough planing 0.5 in thick	Yes	Verify before purchase. Added mass near tongues.
BOM-006	Finish system	1	set	Corrosion-resistant finish or oil/wax coating	18	rough planing 0.5 in thick	Yes	Verify before purchase. Mass and cure can move.
BOM-007	Abrasives and deburring tools	1	set	Files, deburring wheel, abrasive pad, fine abrasive sheet	20	rough planing 0.5 in thick	Yes	Verify before purchase. Must smooth without undulating.
BOM-008	Tuning measurement tools	1	set	Chromatic tuner or Spectrum app, microphone, calipers, feeler block	3	rough planing 0.5 in thick	Yes	Verify before purchase. Required for public-review reading.

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item_id	part_name	required_specs	search_terms	candidate_souces	dates_checked	unit_price	lead_time	substitution_pot	mitigating_risk
SRC-001	Low-carbon steel	1/2" diameter by 12000	Steel light	multiple suppliers	May 06/06	1.00	1 day	Substitute only after knowing grade and measuring	
SRC-002	Steel sheet/top blank	or formed steel	44 gauge mild steel	contacted 6075000	May 06/06	1.00	1 day	Substitute nearby gauge in only gauge easily diff to find	
SRC-003	Cutting service	Laser, waterjet, or CNC	Reserved on good available	check with suppliers	May 06/06	1.00	1 day	Prefer waterjet w/ based on AZ and is acceptable	
SRC-004	Rubber isolation	Soft non-rattling feet with some	decide size and color	May 06/06	1.00	1 day	Substitute only if it does not affect sustain		
SRC-005	Mallets	Medium-soft repeated long	allet pain in side	supplier	May 06/06	1.00	1 day	Substitute only if Hardness less good	
SRC-006	Finish	Low-build corrosion	clear a reaction	set up and test for finishing	May 06/06	1.00	1 day	Substitute thick coating if finish decreases and can dan	

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

cut_id	part_name	qty	material	rough_dimensions	final_dimensions	operation	datum	kerf_allowance	inspection
CUT-001	Shell blank	1	low-carbon steel	12.5 dia x 6.25 hgt	12.0 dia x 6.1 hgt	drill, grind, stoppage	shell line and rim plane	0.040	Confirm roundness, thickness.
CUT-002	Port opening	1	same shell	pilot hole plus trim	2.5 dia port	cut/deburr port	shell centerline	0.040	Measure actual diameter and
CUT-003	Bass tongue slot	1	0.079 in steel top	layout from 4.16	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Leave trim allowance; log first-
CUT-004	Tongue 1 slot	1	0.079 in steel top	layout from 3.39	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Deburr before first measured s
CUT-005	Tongue 2 slot	1	0.079 in steel top	layout from 3.30	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Check kerf and root radius.
CUT-006	Tongue 3 slot	1	0.079 in steel top	layout from 3.11	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Check for heat warp before tun
CUT-007	Tongue 4 slot	1	0.079 in steel top	layout from 2.94	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Measure Hz before material re
CUT-008	Tongue 5 slot	1	0.079 in steel top	layout from 2.77	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Tune in small passes.
CUT-009	Tongue 6 slot	1	0.079 in steel top	layout from 2.69	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Watch for burr mass and damp
CUT-010	Tongue 7 slot	1	0.079 in steel top	layout from 2.54	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Check sustain versus adjacent
CUT-011	Tongue 8 slot	1	0.079 in steel top	layout from 2.40	1.0 tongue	drill, grind, stoppage	top centerline	0.040	A4 formula sanity row.
CUT-012	Tongue 9 slot	1	0.079 in steel top	layout from 2.20	1.0 tongue	drill, grind, stoppage	top centerline	0.040	High tongues may be quiet; ve
CUT-013	Tongue 10 slot	1	0.079 in steel top	layout from 2.08	1.0 tongue	drill, grind, stoppage	top centerline	0.040	Avoid over-thinning short tong
CUT-014	Fixture ring or cradle	1	MDF plywood or printed 1/4" ply	4.0 x 6.0 x 0.75	same flask	hand jig rocking	rim plane	0.000	Must hold shell without dampin

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Drawing Brief

Required Drawing Set

1. Top layout with all tongue slots, centerline, port, rim, and orientation mark.
2. Section view through shell showing top thickness, shell height, port, support feet, and rim/edge treatment.
3. Detail view of a tongue root with slot-end radius, kerf, burr limit, and stress-relief note.
4. Fixture view showing temporary tuning support and final isolation feet.
5. Process coupon drawing for kerf and HAZ validation.

Critical Dimensions

- 12.0 in body diameter assumption.
- 6.0 in shell height assumption.
- 0.079 in steel thickness assumption.
- 0.040 in kerf assumption.
- 1.0 in tongue width assumption.
- 2.5 in port diameter assumption.
- Tongue lengths from `design.md` are derived estimates and must be marked "rough-cut; tune by measurement."

Drawing Rules

- Units: inches with metric equivalents where useful.
- Include title block, revision, date, material, finish, and "not final tuned dimensions" note.
- Show datum A as top center, datum B as rim plane, datum C as orientation line.
- Call out all inside slot ends as smooth radiused transitions.
- Do not publish as final manufacturing drawing until validation data exists.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Assembly Manual

Safety

Steel cutting, deburring, and tuning can create sharp edges, hot work, abrasive dust, and springy tongues. Wear eye protection, gloves when handling raw cuts, and hearing protection during cutting or repeated striking. Do not strike an unfinished blank with sharp burrs in a way that can throw chips.

Tools

- Calipers and steel rule.
- Layout template or CAD print.
- Cutting process: waterjet, laser, CNC mill, or carefully controlled manual process.
- Deburring tools, small files, abrasive pads, and inspection light.
- Tuner or spectrum analyzer with repeatable microphone placement.
- Medium-soft mallets reserved for validation.
- Temporary support cradle and final isolation feet.

Build Sequence

1. Confirm the actual steel grade, thickness, shell diameter, shell height, and port diameter.
2. Update the workbook or `design.md` assumptions if measured thickness differs from 0.079 in.
3. Mark the top centerline, tongue layout, port, and any logo or orientation mark.
4. Cut the port first if it helps workholding; deburr and record actual diameter.
5. Rough-cut tongue slots using rounded slot ends and the documented kerf.
6. Inspect for heat discoloration, slot taper, burrs, and oil-can distortion.
7. Deburr both sides, preserving root radius and avoiding uncontrolled thinning.
8. Place the drum on a temporary isolation support and record first-strike measurements.
9. Tune iteratively, one tongue at a time, recording measured Hz and cents error.
10. Install final feet, edge trim, and finish only after pre-finish tuning is documented.
11. Re-check pitch after final mounting, after finish cure, and after a 24-72 hour drift rest.

Tuning Notes

For this first prototype, do not chase a perfect tuner reading from one strike. Use the same mallet, strike point, support, and microphone distance. Record at least three stable readings per tongue where possible.

If a tongue is sharp after rough cutting, lower pitch by increasing effective length or removing material in a way that reduces stiffness. If a tongue is flat, options are limited; shorten the effective tongue or accept a revised target. This is why rough-cut length should stay conservative.

Photo Checkpoints

- Raw shell and thickness measurement.
- Layout before cutting.
- Slot kerf sample coupon.
- Rough-cut top before deburr.
- Deburred tongue roots.
- First-strike measurement setup.
- Tuning pass with tool marks visible.
- Final mount and underside/support detail.

validation.csv

Target/measured values, tolerance, environment, support, condition, process, state, action, notes

check_id	phase	tongue_or_note	target_hz	measured_hz	charts_err	tolerance	environment	support	condition	process	state	action	notes
VAL-001	formula sanity	A4	A4	440.00			0 cents	shop temp	log temp/humidity	log temp/humidity	log temp/humidity	log temp/humidity	Confirm target formula
VAL-002	rough-cut	bass	D3	146.83			record only	log temp/humidity	medium-soft	after slot cut	open	Measure before deburr if	
VAL-003	rough-cut	tongue 1	A3	220.00			record only	log temp/humidity	medium-soft	after slot cut	open	Do not tune until burrs are	
VAL-004	deburr	all tongues	n/a				no sharp burrs	shop temp	medium-soft	after deburr	open	Inspect both sides and root	
VAL-005	first strike	bass	D3	146.83			plus/minus 50 cents	shop temp	medium-soft	after deburr	open	Record first stable tuner r	
VAL-006	first strike	tongue 1	A3	220.00			plus/minus 50 cents	shop temp	medium-soft	after deburr	open	Repeat for every tongue b	
VAL-007	tuning pass	all tongues	scale targets				final target	shop temp	medium-soft	after deburr	open	Log every trim pass and v	
VAL-008	final mount	all tongues	scale targets				plus/minus 10 cents	shop temp	medium-soft	after deburr	open	Compare mounted reading	
VAL-009	finish check	all tongues	scale targets				plus/minus 10 cents	shop temp	medium-soft	after deburr	open	Check damping and pitch	
VAL-010	drift check	all tongues	scale targets				plus/minus 10 cents	shop temp	medium-soft	after deburr	open	Public-review gate: updat	
VAL-011	structural check	slot roots	n/a				no visible cracks	shop temp	medium-soft	after deburr	open	Inspect root radii under br	
VAL-012	ergonomic check	playing surface	n/a				comfortable shape	shop temp	medium-soft	after deburr	open	Confirm player can strike	

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ Draft

Subject: RFQ - steel tongue drum blank / cutting service

Hello,

I am preparing a prototype steel tongue drum and need either a steel shell/blank, cutting service, or both. The project is a struck musical instrument, so material consistency, low warp, clean edges, and documented kerf matter more than cosmetic speed.

Requested Quote

- Quantity: 1 prototype, with option for 2-3 follow-up blanks.
- Material: low-carbon mild steel preferred for the first prototype.
- Thickness target: near 0.079 in; please quote actual available thickness and tolerance.
- Body target: 12 in diameter, 6 in high shell/vessel or compatible top/shell blank.
- Cutting: tongue slots and 2.5 in port from supplied DXF/SVG when finalized.
- Edge condition: quote as-cut condition, deburring options, and expected kerf/taper.
- Process options: waterjet, fiber laser, CO2 laser, CNC mill, or recommended alternative.

Please Include

- Material grade and data sheet.
- Actual thickness tolerance.
- Cutting kerf estimate.
- Heat affected zone or warp expectations.
- Deburring/edge-finishing options.
- Unit price and lead time.
- Whether nested test coupons can be included for kerf and tuning trials.

Substitutions are acceptable only if the grade, thickness, cutting process, and expected edge condition are clearly identified.

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Visual BOM Brief

Goal

Create a one-page visual BOM that lets a fabricator or reviewer understand the steel tongue drum parts without reading every CSV.

Layout

- Header: Steel Tongue Drum, STD-001, quote date, estimated total.
- Main image: top layout drawing from `drawings/steel-tongue-drum-layout.svg`.
- Row images: shell blank, tongue field, port, rubber feet, mallets, finish, deburring tools.
- Columns: item id, part, visual, quantity, required spec, cost basis, tuning risk.

Required Labels

- "Derived estimate" for any tongue geometry.
- "Verify before purchase" for price and supplier information.
- "Measured data required" near tuning-critical parts.

Shot Priority

Until real photos exist, use clean line drawings and explicit placeholders. Do not use atmospheric concept images as proof of manufactured geometry.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

(* Steel Tongue Drum Wolfram starter *)

```
ClearAll["Global`*"];
```

(* Inputs from steel-tongue-drum-design-table.xlsx *)

```
bodyDiameterIn = 12.0;
```

```
shellHeightIn = 6.0;
```

```
steelThicknessIn = 0.079;
```

```
kerfIn = 0.040;
```

```
tongueWidthIn = 1.0;
```

```
portDiameterIn = 2.5;
```

```
a4Hz = 440.0;
```

```
kSteel = 32098.0;
```

```
targetMidis = <|
```

```
"Bass" -> 50,
```

```
"T1" -> 57,
```

```
"T2" -> 58,
```

```
"T3" -> 60,
```

```
"T4" -> 62,
```

```
"T5" -> 64,
```

```
"T6" -> 65,
```

```
"T7" -> 67,
```

```
"T8" -> 69,
```

```
"T9" -> 72,
```

```
"T10" -> 74
```

```
|>;
```

```
midiToHz[m_] := a4Hz*2^((m - 69)/12);
```

```
tongueLengthIn[f_] := Sqrt[kSteel*steelThicknessIn/f];
```

```
centsError[measured_, target_] := 1200*Log2[measured/target];
```

```
targetTable = KeyValueMap[
```

```
{#1, #2, midiToHz[#2], tongueLengthIn[midiToHz[#2]], 25.4*tongueLengthIn[midiToHz[#2]]} &,
```

```
targetMidis
```

```

];

Grid[
Prepend[targetTable, {"Tongue", "MIDI", "Target Hz", "Length in", "Length mm"}],
Frame -> All
]

(* Validation example: replace with measured data from validation.csv. *)
measuredExampleHz = <|"T8" -> 440.0|>;
KeyValueMap[{#1, centsError[#2, midiToHz[targetMidis[#1]]]} &, measuredExampleHz]

(* Simple sensitivity plot: pitch versus tongue length for the selected steel thickness. *)
Plot[kSteel*steelThicknessIn/l^2, {l, 2.0, 4.5},
AxesLabel -> {"Tongue length (in)", "Frequency (Hz)"},
PlotLabel -> "First-pass cantilever estimate only"
]

```


README.md

Project artifact.

Steel Tongue Drum

Root-mode engineering packet for a tuned steel tongue drum: a slit-tongue steel idiophone whose final tuning must be measured and corrected on the actual prototype.

This repo is part of the [tonykoop/instrument-maker](https://github.com/tonykoop/instrument-maker) catalogue and uses the `instrument-maker-v4` root packet shape.

What this is

This packet turns the existing `steel-tongue-drum-design-table.xlsx` into a build-ready documentation set for a circular steel tongue drum. The workbook provides first-pass geometry, material constants, target MIDI notes, a rough Helmholtz body estimate, and tongue length formulas.

The important caveat: steel tongue drums are empirical instruments. The cantilever equation is useful for rough layout, but final pitch depends on steel batch, residual stress, slot kerf, heat affected zone, burrs, mounting isolation, mallet hardness, finish mass, and the way each tongue is trimmed. This repo is therefore not "tuned" until measured prototype data exists in `validation.csv`.

Current design baseline

Item	Baseline
---	---
Body form	Round steel shell / vessel
Outside diameter	12 in assumption from workbook
Shell height	6 in assumption from workbook
Steel thickness	0.079 in derived from workbook input
Slot kerf	0.040 in workbook assumption
Tongue width	1.0 in workbook assumption
Port diameter	2.5 in workbook assumption
Tuning reference	A4 = 440 Hz
Governing model	$f = K_{\text{steel}} * t / L^2$, first-pass only
Validation state	No measured prototype data yet

Packet map

- `design.md` - design intent, assumptions, formulas, tuning workflow.

- `family-spec.csv` - root-mode family row for the current steel tongue drum.
- `bom.csv`, `sourcing.csv`, `cut-list.csv` - purchasing and shop prep.
- `assembly-manual.md` - rough build, cutting, tuning, finish, and mount steps.
- `validation.csv` - tuning and process validation rows, including rough-cut, deburr, first strike, tuning pass, final mount, and drift check.
- `drawing-brief.md`, `drawings/` - drawing requirements and starter layout SVG.
- `cad/`, `cnc/`, `jigs/` - SolidWorks/CAM/fixture briefs.
- `risks.md` - acoustic, structural, ergonomic, supply, and finish risks with tests.
- `capstone-deck.md`, `print-packet.md`, `print-packet.html`, `site/index.html` - review and presentation layers.

Readiness

This packet is private-prototype ready. It should stay private for public technical review until a rough-cut prototype is struck, tuned, mounted, drift-checked, and logged with measured Hz and cents error.

License

[CC BY 4.0](LICENSE) - see LICENSE for details.

family-spec.csv

Project artifact.

family_id	member_id	instrument	variant	key_or_scale	body_diameter	shell_height	steel_thickness	tongue	cop	pot_diameter	design_stage	validation	state	
STD-FAM-00	STD-001	Steel Tongue	Drum root	Detuned	12 in pentatonic	6 in diatonic	0.079	11	2.5	root-mode	packed	measured	primary	steel-tongue-

jig-decision.md

Project artifact.

Jig Decision

Decision

Use a two-stage jig approach:

1. A non-marring temporary tuning cradle that supports the shell consistently while leaving the tongue field free.
2. A final isolation-foot layout that is re-measured after installation.

Why

The support condition is part of the instrument. A rigid clamp, bench pad, hand hold, rubber foot, and display stand can all change damping and perceived pitch. Separating the temporary tuning cradle from the final mount makes that shift visible in `validation.csv`.

Requirements

- Cradle must locate the shell repeatably without contacting vibrating tongues.
- Contact points should be soft enough to prevent rattles but stable enough for repeatable measurements.
- Fixture should expose the port and underside for inspection.
- Final feet must be installed before the final-mount validation phase.

Rejected Options

- Freehand holding during tuning: too variable.
- Clamping the shell rigidly to a bench: likely to damp shell modes.
- Tuning only after final finish: hides process causes and makes correction harder.

photo-shotlist.md

Project artifact.

Photo Shotlist

- Raw steel shell or blank with label/spec visible.
- Caliper reading of actual steel thickness.
- Layout on top surface before cutting.
- Kerf coupon or test cut close-up.
- Rough-cut slots before deburring.
- Deburred slot root close-up with light showing radius.
- Temporary tuning support setup.
- Tuner/microphone placement during first strike.
- Tuning pass in progress with tool and tongue identified.
- Final feet or isolation mount underside.
- Finished top view with all tongues visible.
- Detail of port, rim, and edge trim.

resources.md

Project artifact.

Resources

Local Packet Sources

- ``steel-tongue-drum-design-table.xlsx`` - workbook source for dimensions, K constants, target notes, and formulas.
- ``design.md`` - narrative design basis and assumptions.
- ``validation.csv`` - required measurement log.
- ``drawings/steel-tongue-drum-layout.svg`` - starter layout drawing.

Instrument Maker References

- ``instrument-maker-v4/references/new-instruments-v4.md`` - steel tongue drum entry.
- ``instrument-maker-v4/references/acoustic-models.md`` - cantilever beam formulas and guard rules.
- ``instrument-maker-v4/references/manufacturing-and-cnc.md`` - machining and fixture guidance.
- ``instrument-maker-v4/references/sourcing-and-production.md`` - sourcing, RFQ, cut list, validation guidance.

Sister Repos

- ``tongue-drum`` - wooden slit tongue drum done-bar reference and DoE pattern.
- ``glockenspiel`` - metal bar idiophone sibling concept.
- ``handpan`` - steel shell idiophone sibling, but different governing model.

Measurement Tools To Use

- Chromatic tuner or spectral analysis app.
- Calipers for thickness, kerf, port, and slot checks.
- Consistent microphone placement.
- Temperature/humidity record for each tuning session.

risks.md

Project artifact.

Risks

Acoustic

| Risk | Why it matters | Verification test |

| --- | --- | --- |

| Cantilever model over-promises accuracy | Real tongues are cut from a coupled steel shell, not ideal beams | First-strike each tongue and log cents error before tuning |

| Mallet hardness changes apparent pitch and partial balance | Hard mallets excite noisy partials and can confuse readings | Record mallet type; compare soft vs medium mallet on two tongues |

| Support condition shifts sustain and pitch | Feet/cradle can damp the shell or tongue roots | Measure before and after final mount |

Structural

| Risk | Why it matters | Verification test |

| --- | --- | --- |

| Slot roots crack | Sharp inside corners concentrate stress | Inspect radiused roots after tuning and after drift check |

| Heat warp from cutting | Warped top changes stiffness and strike height | Photograph and straightedge-check before deburr |

| Over-tuning weakens short tongues | Removing too much material can make tongues unstable | Log every material-removal pass and stop if sustain collapses |

Ergonomic

| Risk | Why it matters | Verification test |

| --- | --- | --- |

| Edge remains sharp | Player contact with rim and port is likely | Glove-off touch test only after full deburr |

| Tongue spacing is cramped | Adjacent tongues may be struck accidentally | Play-test simple scale pattern with final mallets |

Supply

| Risk | Why it matters | Verification test |

| --- | --- | --- |

| Unknown steel grade | K and tuning response vary by material | Record grade/data sheet or mark as unknown in validation |

| Nominal sheet gauge differs from actual | Thickness enters the formula linearly | Caliper thickness at multiple points |

Fit / Finish

| Risk | Why it matters | Verification test |

| --- | --- | --- |

| Finish adds damping or mass | Pitch and sustain can drift after coating | Measure before finish, after cure, and after drift rest |

| Feet or trim rattle | Rattles mask tuning and reduce quality | Strike every tongue at playing volume after final assembly |

validation-report.md

Project artifact.

Validation Report

Current Status

No measured prototype data is present yet. The packet is documentation-complete enough to guide a prototype, but it is not public-review ready as a tuned instrument.

What Has Been Validated

- A4 reference and equal-temperament target method are documented.
- Workbook assumptions have been surfaced in `design.md`.
- The required validation workflow exists in `validation.csv`.
- Manufacturing risks for laser, waterjet, CNC, deburring, support, finish, and drift are documented.

What Remains Open

- Actual steel grade and thickness.
- Actual cutting kerf and edge condition.
- First-strike Hz for every tongue.
- Cents error before and after tuning.
- Pitch shift after final mounting.
- Pitch drift after finish cure and rest.

Readiness Call

Private-prototype ready: yes.

Public-review ready: no. Keep the repo private or clearly marked as unvalidated until measured rough-cut, deburr, first-strike, tuning-pass, final-mount, and drift-check data exists.